Package ‘elements’

September 22, 2025

Title Ecological Niche Models for the Plants of Europe

Description A R package containing Ecological Niche Models (ENMs) for the most prevalent plants, bryophytes and terricolous lichens in the European Vegetation Archive (EVA).

Version 0.5.6

License GPL (>= 3)

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.1

LazyDataCompression xz

Imports e1071 (>= 1.7-16),
filehash (>= 2.4-6)

Suggests knitr,
rmarkdown,
testthat

URL https://github.com/NERC-CEH/elements, https://nerc-ceh.github.io/elements/

BugReports https://github.com/NERC-CEH/elements/issues

Depends R (>= 4.4)

LazyData true

VignetteBuilder knitr

Contents

`ALEData` *ALE data*

Description

A dataset containing the Accumulated Local Effect (ALE) (Molnar, 2018) data for the models, generated using 1000 samples from the training data.

Usage

```
ALEData
```

Format

A data frame with 1514450 rows and 4 columns, the definitions of which are:

taxon_code The taxon, see `elements::TaxonomicBackbone`.

x The variable value.

y The ALE value.

variable The variable name.

Details

```
ALEData
```

`calc_distance` *Calculate the euclidean distance between a set of environmental predictor variables and mutliple taxa*

Description

Calculate the euclidean distance between a set of environmental predictor variables and mutliple taxa

Usage

```
calc_distance(
  predictors,
  taxa = elements::TaxonomicBackbone$taxon_code,
  vars = elements::VariableNames,
  method = "median",
  append = "ids"
)
```

Arguments

`predictors`	A data frame of predictors. Must include the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, bio17, and optionally taxon
`taxa`	A vector of strings containing one or more taxa to calculate distances for. See `elements::TaxonomicBackbone`.
`vars`	A vector of variable names, one or more of `elements::VariableNames`.
`method`	One of "median" or "mean" indicating whether to measure the euclidean distance from the predictor variables to the median or mean values.
`append`	A string, one of "all", "predictors", or "ids" representing which columns from the predictors data frame to return with the results.

Value

A data frame containing the euclidean distance for each taxon.

Examples

```
elements::calc_distance(predictors = elements::ExampleScenarios)
```

`calc_distance_once` *Calculate the euclidean distance between one set of environmental predictor variables and mutliple taxa*

Description

Calculate the euclidean distance between one set of environmental predictor variables and mutliple taxa

Usage

```
calc_distance_once(
  predictors,
  taxa = elements::TaxonomicBackbone$taxon_code,
  vars = elements::VariableNames,
  method = "median",
  append = "ids"
)
```

Arguments

`predictors`	A data frame of predictors. Must include the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, bio17, and optionally taxon
`taxa`	A vector of strings containing one or more taxa to calculate distances for. See `elements::TaxonomicBackbone`.
`vars`	A vector of variable names, one or more of `elements::VariableNames`.
`method`	One of "median" or "mean" indicating whether to measure the euclidean distance from the predictor variables to the median or mean values.
`append`	A string, one of "all", "predictors", or "ids" representing which columns from the predictors data frame to return with the results.

Value

A data frame containing the euclidean distance for each taxon.

Examples

```
elements::calc_distance_once(predictors = elements::ExampleScenarios[1,])
```

`envelope_filter` *Apply an envelope model for a set of predictors for multiple taxa*

Description

Append a column to set of predictors which indicates whether the predictors in each row are within a set of distributional limits for a particular taxon and set of variables.

Usage

```
envelope_filter(
  taxa,
  predictors,
  vars = elements::VariableNames,
  limit = "min_max"
)
```

Arguments

`taxa`	A vector of strings containing one or more taxon codes. See `elements::TaxonomicBackbone`. Optional
`predictors`	A data frame of predictors. Must include one or more of the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, and bio17
`vars`	A vector of strings, containing one or more variables as present in `elements::VariableNames`, which must also be present in the predictors data frame.
`limit`	A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q10_q90", "q25_q75".

Details

This function acts as an envelope, allowing the screening of large quantities of predictor data for the suitability of multiple taxa before applying the more computationally expensive SVM models in `elements::predict_occ` and `elements::predict_occ_taxon`.

Value

The predictors data frame containing an additional column 'within_limit' indicating whether each row of predictord is wthin the specified limits.

Examples

```
## Not run:
elements::envelope_filter_taxon(taxa = NULL, predictors = elements::ExampleData2)

## End(Not run)
```

`envelope_filter_taxon` *Apply an envelope model for a set of predictors for one taxon*

Description

Append a column to set of predictors which indicates whether the predictors in each row are within a set of distributional limits for a particular taxon and set of variables.

Usage

```
envelope_filter_taxon(
  taxon,
  predictors,
  vars = elements::VariableNames,
  limit = "min_max"
)
```

Arguments

- `taxon` — The taxon_code, see `elements::TaxonomicBackbone`.
- `predictors` — A data frame of predictors. Must include one or more of the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, and bio17
- `vars` — A vector of strings, containing one or more variables as present in `elements::VariableNames`, which must also be present in the predictors data frame.
- `limit` — A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q10_q90", "q25_q75".

Details

This function acts as an envelope, allowing the screening of large quantities of predictor data for the suitability of a taxon before applying the more computationally expensive SVM models in `elements::predict_occ` function and to a lesser extent the euclidean distance `elements::calc_distance` function.

Value

The predictors data frame containing an additional column 'within_limit' indicating whether each row of predictord is wthin the specified limits.

Examples

```
## Not run:
elements::envelope_filter(taxon = "stellaria_graminea", predictors = elements::ExampleData1)

## End(Not run)
```

---

env_filter — *Retrieve the most suitable taxa for a given set of environmental variable values*

Description

Retrieve the most suitable taxa for a given set of environmental variable values supplied in the 'predictors' argument. To sets of methods are available:

1. "svm" which generates predictions using `elements::predict_occ` and uses the resultant probability values;
2. "mean" and "median" which calculates the scaled euclidean distance between the values supplied in the 'predictors' argument and the mean or median niche positions as present in `elements::NicheWidths`.

Usage

```
env_filter(
  predictors,
  taxa = elements::TaxonomicBackbone$taxon_code,
  screen = TRUE,
  method = "svm",
  limit = "min_max",
```

```
  exclude = NULL,
  threshold = NULL,
  append = "ids"
)
```

Arguments

`predictors` A data frame of predictors. Must include atleast one the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, bio17. Columns not included must then be included in the 'exclude' argument.

`taxa` A vector of strings containing one or more taxa to generate predictions for.

`screen` A boolean (TRUE/FALSE) indicating whether to use the `elements::envelope_filter` function to check whether taxa are within distibutional limits prior to applying the more computationally expensive `elements::predict_occ` or `elements::calc_distance` functions.

`method` One of "svm", "mean", or "median".

`limit` A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q25_q75". Which if set assigns a probability of 0 to a set of predictors if one or more of those predictors are outside the stipulated quantile ranges. Only applied if pa = "Present". Optional.

`exclude` Model variables to exclude from the distance calculation; passed to the 'holdout' argument of 'elements::predict_occ' if the 'method' argument is "svm", otherwise when the 'method' argument is set to "mean" or "median" those variables are removed from the distance calculation.

`threshold` A probability threshold to use as a cut off in the environmental filter. Only applicable when 'method' = "svm".

`append` A string, one of "all", "predictors", or "ids" representing which columns from the predictors data frame to return with the results.

Details

The svm method will produce more accurate results as it considers the position of the environmental variable values in the 11-dimensional hypervolume; however, if there are a large number of taxa-predictor combinations the mean and median methods offer a faster alternative.

NOTE: The "mean" and "median" methods do not produce realistic results and so are currently included for demonstrative purposes only.

Value

A dataframe containing three columns: taxon_code, rank, and Present (if 'method' = "svm") or Distance (if 'method' = "mean" or "median").

Examples

```
elements::startup(); elements::env_filter(predictors = elements::ExampleScenarios, taxa = elements::Taxonomic
```

EUNISConstantTaxa *EUNISConstantTaxa*

Description

A tibble containing the constant taxa associated with each EUNIS habitat.

Usage

```
EUNISConstantTaxa
```

Format

A tibble with 9108 rows and 5 columns.

eunis_code The EUNIS habitat code.

eunis_name The EUNIS habitat name.

taxon_name The name of the taxon as it appears in Chytrý et al (2020).

taxon_code The taxon code as it appears in `elements::TaxonomicBackbone`.

perc_occ_freq The percentage occurrence frequency of the taxon in the EUNIS habitat.

Details

EUNISConstantTaxa

References

Chytrý, M., Tichý, L., Hennekens, S.M., Knollová, I., Janssen, J.A.M., Rodwell, J.S., Peterka, T., Marcenò, C., Landucci, F., Danihelka, J., Hájek, M., Dengler, J., Novák, P., Zukal, D., Jiménez-Alfaro, B., Mucina, L., Abdulhak, S., Aćić, S., Agrillo, E., Attorre, F., Bergmeier, E., Biurrun, I., Boch, S., Bölöni, J., Bonari, G., Braslavskaya, T., Bruelheide, H., Campos, J.A., Čarni, A., Casella, L., Ćuk, M., Ćušterevska, R., De Bie, E., Delbosc, P., Demina, O., Didukh, Y., Dítě, D., Dziuba, T., Ewald, J., Gavilán, R.G., Gégout, J.-C., Giusso del Galdo, G.P., Golub, V., Goncharova, N., Goral, F., Graf, U., Indreica, A., Isermann, M., Jandt, U., Jansen, F., Jansen, J., Jašková, A., Jiroušek, M., Kącki, Z., Kalníková, V., Kavgacı, A., Khanina, L., Yu. Korolyuk, A., Kozhevnikova, M., Kuzemko, A., Küzmič, F., Kuznetsov, O.L., Laiviņš, M., Lavrinenko, I., Lavrinenko, O., Lebedeva, M., Lososová, Z., Lysenko, T., Maciejewski, L., Mardari, C., Marinšek, A., Napreenko, M.G., Onyshchenko, V., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Rodríguez Rojo, M.P., Rūsiņa, S., Schrautzer, J., Šibík, J., Šilc, U., Škvorc, Ž., Smagin, V.A., Stančić, Z., Stanisci, A., Tikhonova, E., Tonteri, T., Uogintas, D., Valachovič, M., Vassilev, K., Vynokurov, D., Willner, W., Yamalov, S., Evans, D., Palitzsch Lund, M., Spyropoulou, R., Tryfon, E., Schaminée, J.H.J., 2020. EUNIS Habitat Classification: Expert system, characteristic species combinations and distribution maps of European habitats. Applied Vegetation Science 23, 648–675. https://doi.org/10.1111/avsc.12519

`EUNISDiagnosticTaxa` *EUNISDiagnosticTaxa*

Description

A tibble containing the diagnostic taxa associated with each EUNIS habitat.

Usage

```
EUNISDiagnosticTaxa
```

Format

A tibble with 5122 rows and 5 columns.

eunis_code The EUNIS habitat code.

eunis_name The EUNIS habitat name.

taxon_name The name of the taxon as it appears in Chytrý et al (2020).

taxon_code The taxon code as it appears in `elements::TaxonomicBackbone`.

phi_100 The phi coefficient of association between the taxon and EUNIS habitat, a measure of fidelity.

Details

`EUNISDiagnosticTaxa`

References

Chytrý, M., Tichý, L., Hennekens, S.M., Knollová, I., Janssen, J.A.M., Rodwell, J.S., Peterka, T., Marcenò, C., Landucci, F., Danihelka, J., Hájek, M., Dengler, J., Novák, P., Zukal, D., Jiménez-Alfaro, B., Mucina, L., Abdulhak, S., Aćić, S., Agrillo, E., Attorre, F., Bergmeier, E., Biurrun, I., Boch, S., Bölöni, J., Bonari, G., Braslavskaya, T., Bruelheide, H., Campos, J.A., Čarni, A., Casella, L., Ćuk, M., Ćušterevska, R., De Bie, E., Delbosc, P., Demina, O., Didukh, Y., Dítě, D., Dziuba, T., Ewald, J., Gavilán, R.G., Gégout, J.-C., Giusso del Galdo, G.P., Golub, V., Goncharova, N., Goral, F., Graf, U., Indreica, A., Isermann, M., Jandt, U., Jansen, F., Jansen, J., Jašková, A., Jiroušek, M., Kącki, Z., Kalníková, V., Kavgacı, A., Khanina, L., Yu. Korolyuk, A., Kozhevnikova, M., Kuzemko, A., Küzmič, F., Kuznetsov, O.L., Laiviņš, M., Lavrinenko, I., Lavrinenko, O., Lebedeva, M., Lososová, Z., Lysenko, T., Maciejewski, L., Mardari, C., Marinšek, A., Napreenko, M.G., Onyshchenko, V., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Rodríguez Rojo, M.P., Rūsiņa, S., Schrautzer, J., Šibík, J., Šilc, U., Škvorc, Ž., Smagin, V.A., Stančić, Z., Stanisci, A., Tikhonova, E., Tonteri, T., Uogintas, D., Valachovič, M., Vassilev, K., Vynokurov, D., Willner, W., Yamalov, S., Evans, D., Palitzsch Lund, M., Spyropoulou, R., Tryfon, E., Schaminée, J.H.J., 2020. EUNIS Habitat Classification: Expert system, characteristic species combinations and distribution maps of European habitats. Applied Vegetation Science 23, 648–675. https://doi.org/10.1111/avsc.12519

`EUNISDominantTaxa` *EUNISDominantTaxa*

Description

A tibble containing the constant taxa associated with each EUNIS habitat.

Usage

```
EUNISDominantTaxa
```

Format

A tibble with 5122 rows and 5 columns.

eunis_code The EUNIS habitat code.

eunis_name The EUNIS habitat name.

taxon_name The name of the taxon as it appears in Chytrý et al (2020).

taxon_code The taxon code as it appears in `elements::TaxonomicBackbone`.

perc_freq_dom The percentage frequency of plots in which the species occurs with a cover larger than 25%.

Details

```
EUNISDominantTaxa
```

References

Chytrý, M., Tichý, L., Hennekens, S.M., Knollová, I., Janssen, J.A.M., Rodwell, J.S., Peterka, T., Marcenò, C., Landucci, F., Danihelka, J., Hájek, M., Dengler, J., Novák, P., Zukal, D., Jiménez-Alfaro, B., Mucina, L., Abdulhak, S., Aćić, S., Agrillo, E., Attorre, F., Bergmeier, E., Biurrun, I., Boch, S., Bölöni, J., Bonari, G., Braslavskaya, T., Bruelheide, H., Campos, J.A., Čarni, A., Casella, L., Ćuk, M., Ćušterevska, R., De Bie, E., Delbosc, P., Demina, O., Didukh, Y., Dítě, D., Dziuba, T., Ewald, J., Gavilán, R.G., Gégout, J.-C., Giusso del Galdo, G.P., Golub, V., Goncharova, N., Goral, F., Graf, U., Indreica, A., Isermann, M., Jandt, U., Jansen, F., Jansen, J., Jašková, A., Jiroušek, M., Kącki, Z., Kalníková, V., Kavgacı, A., Khanina, L., Yu. Korolyuk, A., Kozhevnikova, M., Kuzemko, A., Küzmič, F., Kuznetsov, O.L., Laiviņš, M., Lavrinenko, I., Lavrinenko, O., Lebedeva, M., Lososová, Z., Lysenko, T., Maciejewski, L., Mardari, C., Marinšek, A., Napreenko, M.G., Onyshchenko, V., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Rodríguez Rojo, M.P., Rūsiņa, S., Schrautzer, J., Šibík, J., Šilc, U., Škvorc, Ž., Smagin, V.A., Stančić, Z., Stanisci, A., Tikhonova, E., Tonteri, T., Uogintas, D., Valachovič, M., Vassilev, K., Vynokurov, D., Willner, W., Yamalov, S., Evans, D., Palitzsch Lund, M., Spyropoulou, R., Tryfon, E., Schaminée, J.H.J., 2020. EUNIS Habitat Classification: Expert system, characteristic species combinations and distribution maps of European habitats. Applied Vegetation Science 23, 648–675. https://doi.org/10.1111/avsc.12519

ExampleData1 *Example predictor data*

Description

A dataset containing a randomised sample of 100 presences and 100 absences from the training and test data for one taxon: Stellaria graminea (stellaria_graminea)

Usage

```
ExampleData1
```

Format

A data frame with 200 rows and 11 columns, the definitions of which are:

L Light

M Soil Moisture

N Soil Nitrogen

R Reaction

S Salinity

SD Soil Disturbance

GP Grazing Pressure

bio05 Maximum temperature in the warmest month

bio06 Minimum temperature in the coldest month

bio16 Precipitation in the wettest quarter

bio17 Precipitation in the driest quarter

Details

ExampleData1

References

Copernicus Climate Change Service, 2021. Downscaled bioclimatic indicators for selected regions from 1950 to 2100 derived from climate projections. https://doi.org/10.24381/CDS.0AB27596

Dengler, J., Jansen, F., Chusova, O., Hüllbusch, E., Nobis, M.P., Meerbeek, K.V., Axmanová, I., Bruun, H.H., Chytrý, M., Guarino, R., Karrer, G., Moeys, K., Raus, T., Steinbauer, M.J., Tichý, L., Tyler, T., Batsatsashvili, K., Bita-Nicolae, C., Didukh, Y., Diekmann, M., Englisch, T., Fernández-Pascual, E., Frank, D., Graf, U., Hájek, M., Jelaska, S.D., Jiménez-Alfaro, B., Julve, P., Nakhutsrishvili, G., Ozinga, W.A., Ruprecht, E.-K., Šilc, U., Theurillat, J.-P., Gillet, F., 2023. Ecological Indicator Values for Europe (EIVE) 1.0. Vegetation Classification and Survey 4, 7–29. https://doi.org/10.3897/VCS.98324

Midolo, G., Herben, T., Axmanová, I., Marcenò, C., Pätsch, R., Bruelheide, H., Karger, D.N., Aćić, S., Bergamini, A., Bergmeier, E., Biurrun, I., Bonari, G., Čarni, A., Chiarucci, A., De Sanctis, M., Demina, O., Dengler, J., Dziuba, T., Fanelli, G., Garbolino, E., Giusso del Galdo, G., Goral, F., Güler, B., Hinojos-Mendoza, G., Jansen, F., Jiménez-Alfaro, B., Lengyel, A., Lenoir, J., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Ruprecht, E., Rūsiņa, S., Šilc, U., Škvorc, Ž., Stančić, Z., Tatarenko, I., Chytrý, M., 2023. Disturbance indicator values for European plants. Global Ecology and Biogeography 32, 24–34. https://doi.org/10.1111/geb.13603

Tichý, L., Axmanová, I., Dengler, J., Guarino, R., Jansen, F., Midolo, G., Nobis, M.P., Van Meerbeek, K., Aćić, S., Attorre, F., Bergmeier, E., Biurrun, I., Bonari, G., Bruelheide, H., Campos, J.A., Čarni, A., Chiarucci, A., Ćuk, M., Ćušterevska, R., Didukh, Y., Dítě, D., Dítě, Z., Dziuba, T., Fanelli, G., Fernández-Pascual, E., Garbolino, E., Gavilán, R.G., Gégout, J.-C., Graf, U., Güler, B., Hájek, M., Hennekens, S.M., Jandt, U., Jašková, A., Jiménez-Alfaro, B., Julve, P., Kambach, S., Karger, D.N., Karrer, G., Kavgacı, A., Knollová, I., Kuzemko, A., Küzmič, F., Landucci, F., Lengyel, A., Lenoir, J., Marcenò, C., Moeslund, J.E., Novák, P., Pérez-Haase, A., Peterka, T., Pielech, R., Pignatti, A., Rašomavičius, V., Rūsiņa, S., Saatkamp, A., Šilc, U., Škvorc, Ž., Theurillat, J.-P., Wohlgemuth, T., Chytrý, M., 2023. Ellenberg-type indicator values for European vascular plant species. Journal of Vegetation Science 34, e13168. https://doi.org/10.1111/jvs.13168

`ExampleData2` *Example predictor data*

Description

A dataset containing a randomised sample of 100 presences and 100 absences from the training and test data for two taxa: Stellaria graminea (stellaria_graminea) and Silene flos-cuculi (silene_flos-cuculi).

Usage

```
ExampleData2
```

Format

A data frame with 400 rows and 12 columns, the definitions of which are:

L Light

M Soil Moisture

N Soil Nitrogen

R Reaction

S Salinity

SD Soil Disturbance

GP Grazing Pressure

bio05 Maximum temperature in the warmest month

bio06 Minimum temperature in the coldest month

bio16 Precipitation in the wettest quarter

bio17 Precipitation in the driest quarter

taxon_code The taxon, see `elements::TaxonomicBackbone`.

Details

```
ExampleData2
```

References

Copernicus Climate Change Service, 2021. Downscaled bioclimatic indicators for selected regions from 1950 to 2100 derived from climate projections. https://doi.org/10.24381/CDS.0AB27596

Dengler, J., Jansen, F., Chusova, O., Hüllbusch, E., Nobis, M.P., Meerbeek, K.V., Axmanová, I., Bruun, H.H., Chytrý, M., Guarino, R., Karrer, G., Moeys, K., Raus, T., Steinbauer, M.J., Tichý, L., Tyler, T., Batsatsashvili, K., Bita-Nicolae, C., Didukh, Y., Diekmann, M., Englisch, T., Fernández-Pascual, E., Frank, D., Graf, U., Hájek, M., Jelaska, S.D., Jiménez-Alfaro, B., Julve, P., Nakhutsrishvili, G., Ozinga, W.A., Ruprecht, E.-K., Šilc, U., Theurillat, J.-P., Gillet, F., 2023. Ecological Indicator Values for Europe (EIVE) 1.0. Vegetation Classification and Survey 4, 7–29. https://doi.org/10.3897/VCS.98324

Midolo, G., Herben, T., Axmanová, I., Marcenò, C., Pätsch, R., Bruelheide, H., Karger, D.N., Aćić, S., Bergamini, A., Bergmeier, E., Biurrun, I., Bonari, G., Čarni, A., Chiarucci, A., De Sanctis, M., Demina, O., Dengler, J., Dziuba, T., Fanelli, G., Garbolino, E., Giusso del Galdo, G., Goral, F., Güler, B., Hinojos-Mendoza, G., Jansen, F., Jiménez-Alfaro, B., Lengyel, A., Lenoir, J., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Ruprecht, E., Rūsiņa, S., Šilc, U., Škvorc, Ž., Stančić, Z., Tatarenko, I., Chytrý, M., 2023. Disturbance indicator values for European plants. Global Ecology and Biogeography 32, 24–34. https://doi.org/10.1111/geb.13603

Tichý, L., Axmanová, I., Dengler, J., Guarino, R., Jansen, F., Midolo, G., Nobis, M.P., Van Meerbeek, K., Aćić, S., Attorre, F., Bergmeier, E., Biurrun, I., Bonari, G., Bruelheide, H., Campos, J.A., Čarni, A., Chiarucci, A., Ćuk, M., Ćušterevska, R., Didukh, Y., Dítě, D., Dítě, Z., Dziuba, T., Fanelli, G., Fernández-Pascual, E., Garbolino, E., Gavilán, R.G., Gégout, J.-C., Graf, U., Güler, B., Hájek, M., Hennekens, S.M., Jandt, U., Jašková, A., Jiménez-Alfaro, B., Julve, P., Kambach, S., Karger, D.N., Karrer, G., Kavgacı, A., Knollová, I., Kuzemko, A., Küzmič, F., Landucci, F., Lengyel, A., Lenoir, J., Marcenò, C., Moeslund, J.E., Novák, P., Pérez-Haase, A., Peterka, T., Pielech, R., Pignatti, A., Rašomavičius, V., Rūsiņa, S., Saatkamp, A., Šilc, U., Škvorc, Ž., Theurillat, J.-P., Wohlgemuth, T., Chytrý, M., 2023. Ellenberg-type indicator values for European vascular plant species. Journal of Vegetation Science 34, e13168. https://doi.org/10.1111/jvs.13168

`ExamplePlot` *The EVA plot used to form the example scenarios*

Description

A dataset containing the taxon presences and percentage cover for the EVA plot used to form the example scenarios present in `elements::ExampleScenarios`.

Usage

```
ExamplePlot
```

Format

A data frame with 6 rows and 3 columns, the definitions of which are:

taxon_name The scientific names of the taxa present in the example plot.

taxon_code The taxon codes for the taxon_name values, see `elements::TaxonomicBackbone`.

cover_perc The percentage cover of the taxa present in example plot.

Details

```
ExamplePlot
```

`ExampleScenarios` *Predictor data for three example scenarios*

Description

A dataset containing predictor data for five example scenarios, in three groups: (A) Climate Change - RCP4.5 and (A) Climate Change - RCP8.5, (B) Grazing Intensification and (B) Grazing Reduction, abd (C) Nutrient Enrichment.

Usage

```
ExampleScenarios
```

Format

A data frame with 26 rows and 14 columns, the definitions of which are:

L Light

M Soil Moisture

N Soil Nitrogen

R Reaction

S Salinity

SD Soil Disturbance

GP Grazing Pressure

bio05 Maximum temperature in the warmest month

bio06 Minimum temperature in the coldest month

bio16 Precipitation in the wettest quarter

bio17 Precipitation in the driest quarter

scenario The scenario name.

timeslice The scenario timeslice, either a year or period.

scenario_code The scenario code: a, b, or c.

Details

```
ExampleScenarios
```

`Imbalances` *Presence-Absence Imbalances*

Description

The number of presences and absences in the EVA, suitable samples within the EVA, and the training and test data, along with the presence-absence and absence-presence imbalance ratios.

Usage

```
Imbalances
```

Format

A data frame with 19893 rows and 6 columns, the definitions of which are:

Stage The plot selection 'stage'.

Absent The number of absences.

Present The number or presences.

PA_Imbalance The presence-absence imbalance ratio.

AP_Imbalance The absence-presence imbalance ratio.

taxon_code The taxon code, see `elements::TaxonomicBackbone`

Details

```
Imbalances
```

`NicheWidths` *Niche width data*

Description

A dataset Niche width data for the modelled taxa.

Usage

```
NicheWidths
```

Format

A data frame with 72941 rows and 14, the definitions of which are:

variable The variable name.

mean The mean variable value.

min The minimum variable value.

max The maximum variable value.

median The median variable value.

q01 The 1% quantile value.

q05 The 5% quantile value.

q10 The 10% quantile value.

q25 The 25% quantile value.

q75 The 75% quantile value.

q90 The 90% quantile value.

q95 The 95% quantile value.

q99 The 99% quantile value.

taxon_code The taxon, see `elements::TaxonomicBackbone`.

Details

`NicheWidths`

`PDPData` *PDP data*

Description

A dataset containing the Partial Dependency Profile (PDP) (Molnar, 2018) data for the models, generated using 1000 samples from the training data.

Usage

```
PDPData
```

Format

A data frame with 1458820 rows and 4 columns, the definitions of which are:

taxon_code The taxon, see `elements::TaxonomicBackbone`.

x The variable value.

y The PDP value.

variable The variable name.

Details

`PDPData`

`PerformanceMeasures` *Model performance measures*

Description

A dataset containing a number of performance measures for each model.

Usage

```
PerformanceMeasures
```

Format

A data frame with 6631 rows and 8 columns, the definitions of which are:

taxon_code The taxon, see `elements::TaxonomicBackbone`.

Holdout.PrecisionRecallAreaUnderCurve The Precision-Recall Area Under the Curve (PRAUC), calculated using the random holdout sample test data.

Holdout.Precision The Precision, calculated using the random holdout sample test data.

Holdout.Recall The Recall, calculated using the random holdout sample test data.

Holdout.Sensitivity The Sensitivity, calculated using the random holdout sample test data.

Holdout.Specificity The Specificity, calculated using the random holdout sample test data.

Holdout.BalancedAccuracy The Balanced Accuracy, calculated using the random holdout sample test data.

STCV.BalancedAccuracy The Balanced Accuracy, calculated during the modelling fitting process which used spatio-temporal 10-fold cross-validation.

Details

```
PerformanceMeasures
```

`plot_me` *Plot the marginal effect values for one or more taxa and set of variables*

Description

Plot the Accumulated Local Effect (ALE) or Partial Dependence Profile (PDP) (Molnar, 2018; Molnar, 2022) marginal effect curves for one or more taxa and a set of variables.

Usage

```
plot_me(
  taxa,
  me_type = "pdp",
  free_y = TRUE,
  presences = TRUE,
  eivs = TRUE,
  normalise = TRUE,
  vars = c("L", "M", "N", "R", "S", "SD", "GP", "bio05", "bio06", "bio16", "bio17"),
  lmw = 15,
  lts = 0.75
)
```

Arguments

`taxa`	A vector of one or more taxon_code strings, see `elements::TaxonomicBackbone`.
`me_type`	A string representing the marginal effect plot type, one of "ale" or "pdp".
`free_y`	A boolean. If TRUE the Y axis scales are independent and free for all subplots. If FALSE the Y axis scales are fixed between all subplots.
`presences`	A boolean. If TRUE a box and whiskers plot showing the distribution of presences along each variable will be displayed.
`eivs`	A boolean. If TRUE a point representing the EIV value and arrows representing the EIV niche widths for the taxon will be displayed, where available in `elements::VariableData`.
`normalise`	A boolean. If TRUE and me_type == "pdp" the y axes are normalised using min-max re-scaling.
`vars`	A vector of variables. Must include atleast one of the following columns: "L", "M", "N", "R", "S", "SD", "GP", "bio05", "bio06", "bio16", and "bio17".
`lmw`	The width of the outer margin containing the legend, passed to the "oma" argument of `graphics::par`. Adjust to ensure the legend is given enough room.
`lts`	The size of the legend text, passed to the "cex" argument of `graphics::legend`. Adjust to ensure the legend text size is appropriate.

Details

If the number of taxa is one, setting the 'presences' argument to TRUE a box and whiskers plot showing the distribution of presences is overlaid and by setting the 'eivs' argument to TRUE a point and arrows showing the EIV and niche width values are overlaid, where available in `elements::VariableData`.

The presence-absence imbalance in the training data varies by taxon. This 'ghost of imbalance' (Jiménez-Valverde & Lobo, 2006) has several impacts on the PDP plots:

1. the optimum value of the PDP curve may be less than 1.
2. the entire PDP curve may sit below y = 0.5 (the presence-absence threshold).

When inspecting the PDP plots, it is therefore important to pay more attention to the shape of the response, rather than the absolute PDP value. However, by setting the normalise argument to TRUE, the PDP plot data is transformed using min-max normalisation/re-scaling.

In some instances the ALE curves may reflect 'inverted' responses which are not ecologically realistic, this is most often seen in situations where the distribution of presences along a variable gradient is extremely narrow and/or where there is a non-unimodal distribution, which causes extrapolation issues in the ALE calculations. For example, *Gymnocarpium robertianum* has a extremely narrow distribution of plot-mean S values, with a maximum value of 1. In these instances it is important to also visualise the PDP plots, which should then be prioritised when inspecting the shape of the univariate response.

Value

A composite plot showing the marginal effects and optionally the distribution of presences for selected model variables.

References

Jiménez-Valverde, A., Lobo, J.M., 2006. The ghost of unbalanced species distribution data in geographical model predictions. Diversity and Distributions 12, 521–524. https://doi.org/10.1111/j.1366-9516.2006.00267.x

Molnar, C., 2018. iml: An R package for Interpretable Machine Learning. Journal of Open Source Software 3, 786. https://doi.org/10.21105/joss.00786

Molnar, C., 2022. Interpretable Machine Learning: A Guide For Making Black Box Models Explainable. Independently published, Munich, Germany.

Examples

```
elements::plot_me(taxa = "ajuga_reptans", me_type = "ale", free_y = FALSE, presences = TRUE, eivs = TRUE, norma

elements::plot_me(taxa = c("galium_boreale", "galium_sylvaticum", "galium_uliginosum"), me_type = "ale", free
```

---

predict_occ — *Generate predictions for the probability of occurrence for specified taxa*

Description

Generate predictions for the probability of occurrence (presence and/or absence) using a data frame of predictors for a one or more taxa specified in either:

1. the 'taxa' argument, or
2. a column in the predictors data frame named 'taxon_code'

Usage

```
predict_occ(
  taxa,
  predictors,
  pa = "Present",
  limit = NULL,
  holdopt = NULL,
  dp = 3,
  append = "ids"
)
```

Arguments

taxa	A vector of strings containing one or more taxa to generate predictions for. See `elements::TaxonomicBackbone`. Optional.
predictors	A data frame of predictors. Must include the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, bio17, and optionally taxon
pa	One of "Present", "Absent", or c("Present", "Absent").
limit	A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q25_q75". Which if set assigns a probability of 0 to a set of predictors if one or more of those predictors are outside the stipulated quantile ranges. Only applied if pa = "Present". Optional.
holdopt	Hold one or more variables at their optimum values. NULL by default, else a vector of variable codes, e.g. c("SD", "GP").
dp	The number of decimal places to round the probability values to.
append	A string, one of "all", "predictors", or "ids" representing which columns from the predictors data frame to return with the results.

Details

A number of optional arguments provide additional control over the use of the models. First, using the 'limit' argument the probabilities may be set to 0 if one or more of the predictor values are outside a stipulated quantile range, e.g. the 1% and 99% quantiles by setting the 'limit' argument to "q01_q99". This may be used to strictly enforce the assignment of probability values of 0. As the quantile values present in `elements::NicheWidths` are derived using all presences of each taxon in the EVA, they may better represent the upper and lower tolerances of a taxons ecological niche, especially for taxa where the presences were undersampled. Second, using the 'holdopt' argument one or more predictor variables can be held constant, e.g. to hold soil disturbance constant supply c("SD") to the 'holdopt' argument. This may be useful in instances where you wish to remove the influence of one or more variables on the model results.

If taxon codes are supplied in the 'taxa' argument and there is a column in the predictors data frame named 'taxon_code', the taxa present in the 'taxa' argument will be used and results will be calculated using the entire predictors data frame and the 'taxon_code' column will be replaced.

NOTE: to use this function you must first run `elements::startup()`

Value

A data frame containing the probability of occurrence (Present and/or Absent) for each taxon.

Examples

```
## Not run:
elements::startup()

# Generate predictions using a data frame containing taxon_codes in the 'taxon' column.
elements::predict_occ(taxa = NULL, predictors = elements::ExampleData2, pa = "Present", limit = NULL, holdopt =

# Generate predictions for taxa specified in the 'taxa' argument using a data frame containing only predictor va
elements::predict_occ(taxa = c("stellaria_graminea", "silene_flos-cuculi"), predictors = elements::ExampleDat

# Generate predictions for taxa specified in the 'taxa' argument using a data frame containing both predictor va
elements::predict_occ(taxa = c("stellaria_graminea", "silene_flos-cuculi"), predictors = elements::ExampleDat
```

```
elements::shutdown()

## End(Not run)
```

---

predict_occ_taxon *Generate predictions for the probability of occurrence*

Description

Generate predictions for the probability of occurrence (presence and/or absence) for a given taxon.

Usage

```
predict_occ_taxon(
  taxon,
  predictors,
  pa = "Present",
  limit = NULL,
  holdopt = NULL,
  dp = 3,
  append = "ids"
)
```

Arguments

taxon	The taxon_code, see `elements::TaxonomicBackbone`.
predictors	A data frame of predictors. Must include the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, and bio17
pa	One of "Present", "Absent", or c("Present", "Absent").
limit	A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q10_q90", "q25_q75". Which if set assigns a probability of 0 to the Present column and/or 1 to the Absent column to a set of predictors if one or more of those predictors are outside the stipulated quantile ranges. Optional.
holdopt	Hold one or more variables at their optimum values. NULL by default, else a vector of variable codes, e.g. c("SD", "GP").
dp	The number of decimal places to round the probability values to.
append	A string, one of "all", "predictors", or "ids" representing which columns from the predictors data frame to return with the results.

Details

A number of optional arguments provide additional control over the use of the models. First, using the 'limit' argument the probabilities may be set to 0 if one or more of the predictor values are outside a stipulated quantile range, e.g. the 1% and 99% quantiles by setting the 'limit' argument to "q01_q99". This may be used to strictly enforce the assignment of probability values of 0. As the quantile values present in `elements::NicheWidths` are derived using all presences of each taxon in the EVA, they may better represent the upper and lower tolerances of a taxons ecological niche, especially for taxa where the presences were undersampled. Second, using the 'holdopt' argument one or more predictor variables can be held constant, e.g. to hold soil disturbance constant supply c("SD") to the 'holdopt' argument. This may be useful in instances where you wish to remove the influence of one or more variables on the model results.

NOTE: to use this function you must first run `elements::startup()`

Value

A data frame containing the probability of occurrence (Present and/or Absent).

Examples

```
## Not run:
elements::startup()
elements::predict_occ_taxon(taxon = "stellaria_graminea", predictors = elements::ExampleData1)
elements::shutdown()

## End(Not run)
```

`shutdown` *Remove the filehash database connection from the global environment*

Description

Remove the filehash database connection from the global environment

Usage

```
shutdown()
```

Examples

```
elements::shutdown()
```

`startup` *Initiliase a connection to the filehash database containing the Ecological Niche Models (ENMs)*

Description

Initiliase a connection to the filehash database containing the Ecological Niche Models (ENMs)

Usage

```
startup(models = NULL)
```

Arguments

`models`	One of "all", "test", or NULL, indicating whether the complete set of models should be loaded, or the test set of models. NULL by default, which loads all models.

Value

A connection to the filehash database containing the ENMs in an environment named elementsEnv within the parent environment.

Examples

```
elements::startup()
```

`TaxonomicBackbone` *Taxonomic backbone*

Description

The taxon names and codes for the modeled taxa in the EVA, with the associated taxon concept information retrieved from GBIF.

Usage

```
TaxonomicBackbone
```

Format

A data frame with 6631 rows and 13 columns, the definitions of which are:

taxon_name The taxon names for the modelled taxa.

taxon_code The taxon codes used throughout the package, formed from the taxon_name values by coercing all letters to lower and replacing whitespace with underscores.

scientificName The full scientific name of the taxon, which includes the author. Retrieved from GBIF.

canonicalName The full name of the taxon. Retrieved from GBIF.

species The accepted name for the taxon, following GBIF.

genus The parent Genus taxon associated with the Species.

family The parent Family taxon associated with the Species.

order The parent Order taxon associated with the Species.

class The parent Class taxon associated with the Species.

phylum The parent Phylum taxon associated with the Species.

kingdom The parent Kingdom taxon associated with the Species.

Details

```
TaxonomicBackbone
```

`VariableData` *Environmental indicator value data*

Description

A dataset containing a combined dataset of Environmental Indicator Values (EIVs) from Dengler et al (2023), Midolo et al (2023), and Tichý et al (2023).

Usage

```
VariableData
```

Format

A data frame with 9406 rows and 26 columns, the definitions of which are:

DF Disturbance frequency (Midolo et al., 2023)

DF.sd Disturbance frequency standard deviation (Midolo et al., 2023)

DFH.sd Disturbance frequency herb layer standard deviation (Midolo et al., 2023)

DFh Disturbance frequency herb layer (Midolo et al., 2023)

DS Disturbance severity (Midolo et al., 2023)

DS.sd Disturbance severity standard deviation (Midolo et al., 2023)

DSH.sd Disturbance severity herb layer standard deviation (Midolo et al., 2023)

DSh Disturbance severity herb layer (Midolo et al., 2023)

GP Grazing pressure (Midolo et al., 2023)

GP.sd Grazing pressure standard deviation (Midolo et al., 2023)

L Light (Dengler et al., 2023)

L.nw Light niche width (Dengler et al., 2023)

M Moisture (Dengler et al., 2023)

M.nw Moisture niche width (Dengler et al., 2023)

MF Mowing frequency (Midolo et al., 2023)

MF.sd Mowing frequency standard deviation (Midolo et al., 2023)

N Nitrogen (Dengler et al., 2023)

N.nw Nitrogen niche width (Dengler et al., 2023)

R Reaction (Dengler et al., 2023)

R.nw Reaction niche width (Dengler et al., 2023)

S Salinity (Tichý et al 2023)

SD Soil disturbance (Midolo et al., 2023)

SD.sd Soil disturbance standard deviation (Midolo et al., 2023)

T Temperature (Dengler et al., 2023)

T.nw Temperature niche width (Dengler et al., 2023)

taxon_name The taxon name.

Details

`VariableData`

References

Copernicus Climate Change Service, 2021. Downscaled bioclimatic indicators for selected regions from 1950 to 2100 derived from climate projections. https://doi.org/10.24381/CDS.0AB27596

Dengler, J., Jansen, F., Chusova, O., Hüllbusch, E., Nobis, M.P., Meerbeek, K.V., Axmanová, I., Bruun, H.H., Chytrý, M., Guarino, R., Karrer, G., Moeys, K., Raus, T., Steinbauer, M.J., Tichý, L., Tyler, T., Batsatsashvili, K., Bita-Nicolae, C., Didukh, Y., Diekmann, M., Englisch, T., Fernández-Pascual, E., Frank, D., Graf, U., Hájek, M., Jelaska, S.D., Jiménez-Alfaro, B., Julve, P., Nakhutsrishvili, G., Ozinga, W.A., Ruprecht, E.-K., Šilc, U., Theurillat, J.-P., Gillet, F., 2023. Ecological Indicator Values for Europe (EIVE) 1.0. Vegetation Classification and Survey 4, 7–29. https://doi.org/10.3897/VCS.98324

Midolo, G., Herben, T., Axmanová, I., Marcenò, C., Pätsch, R., Bruelheide, H., Karger, D.N., Aćić, S., Bergamini, A., Bergmeier, E., Biurrun, I., Bonari, G., Čarni, A., Chiarucci, A., De Sanctis, M., Demina, O., Dengler, J., Dziuba, T., Fanelli, G., Garbolino, E., Giusso del Galdo, G., Goral, F., Güler, B., Hinojos-Mendoza, G., Jansen, F., Jiménez-Alfaro, B., Lengyel, A., Lenoir, J., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Ruprecht, E., Rūsiņa, S., Šilc, U., Škvorc, Ž., Stančić, Z., Tatarenko, I., Chytrý, M., 2023. Disturbance indicator values for European plants. Global Ecology and Biogeography 32, 24–34. https://doi.org/10.1111/geb.13603

Tichý, L., Axmanová, I., Dengler, J., Guarino, R., Jansen, F., Midolo, G., Nobis, M.P., Van Meerbeek, K., Aćić, S., Attorre, F., Bergmeier, E., Biurrun, I., Bonari, G., Bruelheide, H., Campos, J.A., Čarni, A., Chiarucci, A., Ćuk, M., Ćušterevska, R., Didukh, Y., Dítě, D., Dítě, Z., Dziuba, T., Fanelli, G., Fernández-Pascual, E., Garbolino, E., Gavilán, R.G., Gégout, J.-C., Graf, U., Güler, B., Hájek, M., Hennekens, S.M., Jandt, U., Jašková, A., Jiménez-Alfaro, B., Julve, P., Kambach, S., Karger, D.N., Karrer, G., Kavgacı, A., Knollová, I., Kuzemko, A., Küzmič, F., Landucci, F., Lengyel, A., Lenoir, J., Marcenò, C., Moeslund, J.E., Novák, P., Pérez-Haase, A., Peterka, T., Pielech, R., Pignatti, A., Rašomavičius, V., Rūsiņa, S., Saatkamp, A., Šilc, U., Škvorc, Ž., Theurillat, J.-P., Wohlgemuth, T., Chytrý, M., 2023. Ellenberg-type indicator values for European vascular plant species. Journal of Vegetation Science 34, e13168. https://doi.org/10.1111/jvs.13168

`VariableLookup` *Environmental variable names and codes*

Description

A dataset containing a lookup for the Ecological Indicator Value (EIV) and bioclimatic variable codes used throughout the `elements` package and full variable namesnames.

Usage

```
VariableLookup
```

Format

A data frame with 29 rows and 5 columns, the definitions of which are:

raw_name The raw name of the variable as present in the parent dataset.

variable_code The variable code, used throughout the `elements` R package.

variable_name A 'tidy' variable name without spaces.

variable_plot_name A 'tidy' variable name with spaces.

model_var A boolean indicating whether the variable was included as a predictor.

Details

```
VariableLookup
```

`VariableNames` *Model variables codes*

Description

A vector containing the environmental variable codes for the model variables; equivalent to the variable_code values in the `elements::VariableLookup` data frame for rows where model_var is TRUE.

Usage

```
VariableNames
```

Format

A vector containing strings.

Details

VariableNames

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